

When Packet Length Is Not Packet Length: Correcting CICFlowMeter Features for Interpretable NIDS Evaluation

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Abstract. The evaluation of network intrusion detection systems (NIDS) heavily relies on labelled datasets of network flows. Among these datasets, CIC-IDS 2017 has become one of the most widely used in network intrusion detection, together with other datasets that rely on CICFlowMeter as their flow extraction tool. Over the years, many studies have addressed flaws in CIC-IDS 2017 and CICFlowMeter that could lead to an incorrect assessment of NIDS performance. In this work, we identify additional errors in CICFlowMeter that lead to the incorrect calculation of some features. More specifically, we show that some statistical features calculated by CICFlowMeter do not match the meaning suggested by their names. We correct these issues and release a new version of CICFlowMeter. This version includes both the original feature set, with corrected names for the misnamed features, and an additional set of features that contains the correct values for those features. Additionally, we remove duplicate features. We apply our corrected version of CICFlowMeter to CIC-IDS 2017 and InSDN, another popular intrusion detection dataset, and demonstrate the impact of these corrections on feature importance analysis and alert explainability.

Keywords: Network intrusion detection · Network flow extraction · Dataset.

1 Introduction

In later years, research on network intrusion detection systems (NIDS) has converged towards machine learning-based solutions to overcome the limitations of traditional signature-based approaches. To assess the performance of these solutions, most studies rely on public datasets containing labeled network flows linked to normal and malicious activity over a network. As collecting and labelling these datasets is a costly operation, high-quality network intrusion detection datasets are considered a valuable asset. Among these, CIC-IDS 2017 is probably the most widely used in the existing literature. The dataset contains labelled network flows, extracted with the CICFlowMeter tool, that were collected on a network testbed over the course of 5 days during working ours. The

PCAP files containing captures of the raw network traffic have also been made publicly available, allowing to reproduce the network flow extraction. Due to its popularity, many research works focused their effort on improving CIC-IDS 2017, fixing errors in the labelling or network flow calculation. The latter were often caused by bugs in CICFlowMeter, as detailed in §2.2.

Our work builds on top of the CICFlowMeter version released by Liu *et al* [18], which incorporates most of the previous fixes made to the tool. In this version, we identify and fix additional errors in CICFlowMeter that lead to the incorrect calculation of 18 network flow attributes (henceforth called features, following the machine learning terminology) and to 2 redundant features.

More specifically, the values of these features represent valid flow statistics, but the value expressed by these statistics does not match the feature name. For example, a feature called `Fwd Segment Size Avg` calculates the average L4 payload length, whereas another feature called `Fwd Seg Size Min` calculates the minimum L4 header length. These inconsistencies between feature names and values can hinder the validity of studies that focus on feature importance or alert explainability. We provide a corrected version of CICFlowMeter that fixes the incorrect network flow features in a two-fold manner: on one hand, it preserves the original feature set by assigning the correct name to the misnamed features; on the other hand, it calculates additional new features that reflect the true values of the original feature names provided by the original feature names.

The contributions of this paper can be summarised as follows:

- We release a fork of CICFlowMeter that includes the correction of previously undocumented semantic inconsistencies and redundant features, described in section 3.2.
- We apply our fixed CICFlowMeter to two popular datasets, CIC-IDS 2017 and InSDN, and release the new version of these datasets.
- We evaluate the impact of our fix to the performance of supervised and unsupervised machine learning models, comparing different feature subsets and evaluating the impact of redundant features, as described in sections 4 and 5.

All the code and artefacts used in the paper will be made available upon publication.

2 Background and Related Work

2.1 Network flows and intrusion detection

Modern research on network intrusion detection, particularly approaches based on machine learning (ML), largely relies on network flow analysis to detect attacks or abnormal activity [4]. In this context, a network flow is typically characterised as a tuple:

$$x = (t, \text{Src IP}, \text{Src Port}, \text{Dst IP}, \text{Dst Port}, \text{Protocol}, \dots) \quad (1)$$

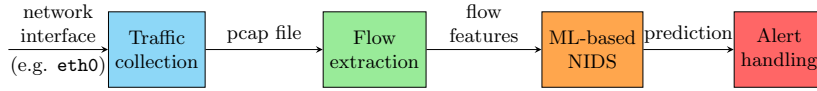


Fig. 1: Typical ML-based NIDS workflow.

where t is the timestamp of the first packet in the bidirectional communication. Tuple 1 describes the minimal information to uniquely characterise a flow, with **Src Port** and **Dst Port** potentially being empty for some protocols (e.g., ICMP).

Building on this representation, the majority of intrusion detection systems in the scientific literature focus on analysing statistical properties of these flows, such as the distribution of packet interarrival times, packet lengths, and TCP flag counts, among others. To extract these features, dedicated flow extraction tools, hereafter called *flowmeters*, are commonly used. Among these, the most popular are CICFlowMeter [7], NetFlow [13], Zeek [20], and NFStream [3].

Figure 1 displays a typical ML-based NIDS workflow. First, raw network traffic is collected from a network interface, and flow-level information is extracted using a flowmeter. Next, an ML-based NIDS labels each network flow as either benign or malicious based on rules learned during training. An ML-based NIDS can then be seen as a function $\mathcal{D}$ mapping a flow x to a prediction $\hat{y} \in \{0, 1\}$ for binary classification (benign/malicious) or $\hat{c} \in \mathbb{Z}_M$ for multiclass classification (one “benign” class and $M - 1$ attack classes).

2.2 CICFlowMeter

CICFlowMeter¹ is one of the most widely used flowmeters in the scientific literature. Many popular intrusion detection datasets such as CIC-IDS 2017 and CSE-CIC-IDS 2018 have been created using this tool. It was originally developed by Lashkari *et al* [7,17] to provide a statistical description of VPN and Tor traffic, and was later extended by Sharafaldin *et al* [26] for intrusion detection.

Over the years, several “bugs” have been discovered in the different datasets produced using CICFlowMeter, part of them being a consequence of errors in the tool. Engelen *et al* were the first to propose fixes for CICFlowMeter, leading to the creation of a forked version of the tool [11]. They fixed the TCP flow termination to prevent the creation of meaningless artefacts, called *TCP appendices*. Additionally, they identified that this bug and TCP timeouts both led to the inversion of the traffic direction, having an impact on the label assignment.

Rosay *et al* [22] further identified issues related to the misinterpretation of ARP packets as IP packets, leading to the extraction of wrong IP addresses and to the creation of wrong flow descriptions. In addition, they identified feature duplication as well as calculation issues on some features and ended up integrating their fixes in LycoSTand [21], an alternative flowmeter.

¹ <https://www.unb.ca/cic/research/applications.html#CICFlowMeter>

Table 1: List of previously identified issues in CICFlowMeter and related fixes. ✓ = fixed in CICFlowMeter; ● = partially fixed in CICFlowMeter; ✕ = not fixed; † = fixed in other tools.

#	Issue	Fixed	Reference(s)
0	TCP flow closed by single FIN flag	✓	[11], [22]
1	TCP flow not terminated by RST flag	✓	[11], [22]
2	Inversion of traffic direction because of TCP time-out	●	[11]
3	SYN-ACK packet captured before SYN packet	✕	[16], [18]
4	TCP segmentation offset issue	✓	[18]
5	Incorrect calculation of existing attributes	✓	[18], [22]
6	ARP packets interpreted as IP packets	†	[22]
7	Feature duplication	†	[22]

Separately, Lanvin *et al* [16] discovered that SYN-ACK packets were surprisingly captured before SYN packets, leading to a debate in the community of CICFlowMeter about the approach for fixing it. Liu *et al* proposed to invert SYN-ACK packets and SYN packets to recreate the logical order of the flow, whereas Lanvin *et al* highlighted that an attacker might deliberately craft a SYN-ACK packet as the first packet of the flow [15]. No proper fix has been mentioned yet in the literature for this issue.

In addition to the bugs already addressed in previous work, Liu *et al* [18] discovered a further issue related to the TCP Segmentation Offset (TSO). They found that CICFlowMeter captured packets before the Network Interface Card (NIC) defined the IP field **Total Length**, leading to the creation of TCP artefacts. Liu *et al* mitigated this issue by fixing the *jNetPcap* library and integrated it in the forked version of Engelen *et al*. We further note that, after publication, the associated repository received additional commits whose impact is not assessed in this work.

2.3 CIC-IDS 2017

CIC-IDS 2017 [26] was created to propose a reliable dataset illustrating real-world criteria for training Machine Learning-based Intrusion Detection Systems. At this time, it was the only dataset to integrate diverse and up-to-date real-world attacks in a testbed involving a wide range of machines running under different operating systems. Indeed, other datasets such as KDD-Cup99 [27] and NSL-KDD [28] existed but suffered from several issues like traffic duplication, deprecated versions of the network stacks and outdated attacks that do not reflect the behaviour of modern networks and the threats they face [10].

Beside the errors discovered across the years in CICFlowMeter and described in Table 1, other issues related to the execution of some attacks and to the labelling strategy have been identified. Table 2 displays a summary of these issues.

² <https://www.wireshark.org/docs/man-pages/reordercap.html>

³ <https://www.wireshark.org/docs/man-pages/editcap.html>

Table 2: Known issues about raw PCAP files, attack execution and labelling process in CIC-IDS 2017.

#	Issue	Related work
1	Failed connections labelled as attacks and adding meaningless noise to the classes (Bot traffic, DoS making the server unable to respond).	Engelen <i>et al</i> (2021) [11]
2	Lack of precision in the timeframes used for traffic labelling.	Engelen <i>et al</i> (2021) [11]
3	Use of deprecated tools to run attacks (DoS Hulk) making them ineffective and creating noise in the dataset.	Engelen <i>et al</i> (2021) [11]
4	Lack of information on the diversity of the configurations of the tools used to run the attacks.	Engelen <i>et al</i> (2021) [11]
5	High-rate traffic induced by DoS and DDoS causes a bad ordering of the packet capture (timestamps are not aligned). Fixed by running <code>reordercap</code> ² on the PCAP files.	Lanvin <i>et al</i> (2022) [16]
6	Packet duplication in raw PCAPs leading to 4.5% of noise in the daily traffic. It has impact on the flow descriptions extracted by CICFlowMeter. Fixed by running <code>editcap</code> ³ on the raw PCAPs.	Lanvin <i>et al</i> (2022) [16]
7	Attack omission for portscan causing labelling issues on Thursday traffic.	Lanvin <i>et al</i> (2022) [16]
8	Labelling errors because of empty payloads caused by the TCP appendices, ports closed preventing the attack to occur, attack startups without malicious content, partitioned flows without malicious payload, attack artefacts which do not illustrate attacks, un-responsive systems following a denial-of-service.	Liu <i>et al</i> (2022) [18]

2.4 InSDN

InSDN [10] was created to propose a testbed environment for intrusion detection tasks in a Software-Defined Network (SDN). These networks are divided into a *control plane*, which dynamically programs the behaviour of devices from the *data plane* such as routers, firewalls, switches, etc. [12]. These two planes introduce new vulnerabilities on the controller itself or on the communication channels between the *control plane* and the *data plane*. InSDN takes into account the vulnerabilities focusing on both the *data plane* and *control plane*. To the best of our knowledge, this dataset is the most widely used for experiments focusing on the detection of intrusions in SDNs. The flows of InSDN have been generated using an old version of CICFlowMeter (most likely v3⁴) and thus suffer from the same issues described in Table 1. Unlike CIC-IDS 2017, and to the best of our knowledge, no mitigation work focusing on InSDN has been reported despite its many citations in the literature [9,24,29].

2.5 Length quantities naming convention

The OSI reference model [6] organises network communication into seven layers. For flow extraction, CICFlowMeter primarily considers two of them: the *network layer*, layer 3 (L3), and the *transport layer*, layer 4 (L4). The unit of communication at each layer is referred to as a *Protocol Data Unit* (PDU): at L3, the PDU is commonly called a “packet”, while for TCP at L4, it is called a “segment”.

⁴ https://github.com/iPAS/TCPDUMP_and_CICFlowMeter/tree/master/CICFlowMeters/CICFlowMeter-3.0

Although the term “segment” is specific to TCP, we use it here as a general term to denote the L4 PDU, regardless of the underlying transport protocol. Each PDU is composed of a header and a payload [1,8]. Based on these definitions, three distinct length quantities can be derived from a given communication:

- Packet length: `L3 header` + `L4 header` + `L4 payload`.
- Packet payload / segment length: `L4 header` + `L4 payload`.
- Segment payload length: `L4 payload`.

3 Fixes

3.1 Issues found in the latest version of CICFlowMeter

We performed a set of sanity checks on the CSV files extracted with Liu *et al.*’s version of CICFlowMeter⁵. In particular, we checked whether statistical features followed the expected relation $\min \leq \text{mean} \leq \max$ and searched for pairs of columns with nearly identical row-wise values. These checks revealed that some packet related features were numerically identical to segment related features, even though they represent different OSI layers (see Section 2.5). This prompted a closer analysis of duplicated columns and inconsistent feature definitions, leading to the identification of the issues described below.

Inconsistent definition of packet length features. The inspection of the source code revealed that all features labelled as packet length were computed using only the segment payloads without the packet or segment header lengths.

Miscomputation of segment length features. On one hand, `Fwd Segment Size Avg` and `Bwd Segment Size Avg` were computed only from the segment payload length (and had the same values as `Fwd Packet Length Mean` and `Bwd Packet Length Mean` respectively), without the segment headers. On the other hand, `Fwd Seg Size Min` and `Bwd Seg Size Min` were computed from the minimum L4 header lengths only, rather than from the entire segment length which should include the segment header and the segment payload.

Redundant features. Redundancies were found between `Packet Length Std` and `Packet Length Variance`, as the variance is the square of the standard deviation; and between `Average Packet Size` and `Packet Length Mean`, which both describe the mean of packet lengths in a flow.

3.2 Corrections and new features

Proper computation of packet and segment length features. We corrected the computation of all packet and segment length features following the definitions given in section 2.5.

⁵ <https://github.com/GintsEngelen/CICFlowMeter>, commit 4dd5319.

Table 3: Feature-level corrections applied to CICFlowMeter. Each row lists an affected feature, the issue in the original computation or naming, the corrected interpretation used in our fixed feature sets, and when applicable, the renamed feature under which the legacy value is preserved.

Affected feature	Original value/issue	Corrected value	Legacy value preserved as
Total Length of Fwd Packet	L4 payload only	Actual packet length	Fwd Segment Payload Length Total
Total Length of Bwd Packet	L4 payload only	Actual packet length	Bwd Segment Payload Length Total
Fwd Packet Length Max	L4 payload only	Actual packet length	Fwd Segment Payload Length Max
Fwd Packet Length Min	L4 payload only	Actual packet length	Fwd Segment Payload Length Min
Fwd Packet Length Mean	L4 payload only	Actual packet length	Fwd Segment Payload Length Mean
Fwd Packet Length Std	L4 payload only	Actual packet length	Fwd Segment Payload Length Std
Bwd Packet Length Max	L4 payload only	Actual packet length	Bwd Segment Payload Length Max
Bwd Packet Length Min	L4 payload only	Actual packet length	Bwd Segment Payload Length Min
Bwd Packet Length Mean	L4 payload only	Actual packet length	Bwd Segment Payload Length Mean
Bwd Packet Length Std	L4 payload only	Actual packet length	Bwd Segment Payload Length Std
Packet Length Max	L4 payload only	Actual packet length	Segment Payload Length Max
Packet Length Min	L4 payload only	Actual packet length	Segment Payload Length Min
Packet Length Mean	L4 payload only	Actual packet length	Segment Payload Length Mean
Packet Length Std	L4 payload only	Actual packet length	Segment Payload Length Std
Fwd Segment Size Avg	L4 payload only; duplicate of Fwd Packet Length Mean	Actual segment length	—
Bwd Segment Size Avg	L4 payload only; duplicate of Bwd Packet Length Mean	Actual segment length	—
Fwd Seg Size Min	Fwd L4 header length min (unrelated to segment)	Actual segment length	Fwd Segment Header Length Min
Bwd Seg Size Min	Bwd L4 header length min (unrelated to segment)	Actual segment length	Bwd Segment Header Length Min
Packet Length Variance	Redundant with Packet Length Std	Removed	—
Average Packet Size	Semantically and numerically identical to Packet Length Mean	Removed	—

Preservation of previously captured information. Since several packet length features previously captured only the L4 payload length, their computation was corrected while the former payload-only values were preserved under explicit **Segment Payload Length** feature names. Counterparts were added only when they preserved non-redundant information; in particular, no separate counterparts were introduced for **Fwd/Bwd Segment Size Avg**, as these values were already covered by the corresponding **Fwd/Bwd Segment Payload Length Mean** features. Finally, **Packet Length Variance** and **Average Packet Size** were removed to avoid feature duplication. The full set of corrections, additions, and removals is summarised in Table 3. The new complete list of features will be disclosed on our GitHub page upon publication.

4 Experimental methodology

The fixes introduced by this paper naturally raise a number of questions, which can be summarised as follows:

RQ1: *Do the corrected feature semantics help to achieve significantly better results for intrusion detection tasks?*

RQ2: *Does the presence of duplicate/redundant features have a significant effect on the results?*

RQ3: *Do the fixes affect the interpretation of the feature importance and, consequently, the conclusions drawn about the model behaviour?*

In this section, we illustrate the methodology used to empirically answer these questions by evaluating the impact of different feature subsets on the performance and result interpretability of ML-based NIDS.

4.1 NIDS models

In our experiments, we consider two types of ML-based NIDS, which we denote by $\mathcal{D}_{\text{RF}}$ and $\mathcal{D}_{\text{AE}}$, respectively.

Supervised classifier $\mathcal{D}_{\text{RF}}$. A Random Forest classifier, which is a supervised model and therefore requires labelled data for training. For this model, we employ the same hyperparameter configuration as Apruzzese *et al.* [4] and use the same 80/20 split⁶. While tree-based classifiers such as Random Forests are typically considered “shallow” compared to modern neural network models, they exhibit strong performance on supervised network intrusion detection tasks.

Anomaly detector $\mathcal{D}_{\text{AE}}$. An AutoEncoder employed as an anomaly detector. The AutoEncoder learns to compress and reconstruct the numerical features of network flows observed during training. As a result, it can accurately reconstruct normal (benign) traffic, while anomalous traffic is reconstructed less accurately. At inference time, a reconstruction error is computed for each flow as the L^2 distance between the original flow and its reconstruction. This error is then compared against a threshold to classify flows as “normal” or “abnormal”. The threshold is typically determined using heuristics, for example by selecting a high percentile of the reconstruction error computed on normal traffic, a process we refer to as calibration. In our experimental evaluation, we use the same AutoEncoder architecture and hyperparameters as [2] and the same split.

4.2 Flow labelling

For CIC-IDS 2017, we adopt the corrected labels provided by Liu *et al* [18], who identified and resolved a number of labelling inconsistencies in the original dataset. Rather than re-running their labelling script, which did not reproduce the labels from their published CSVs when applied to flows extracted with their tool⁵, we use the labelled CSV files they published alongside their paper for reference. Labels are transferred via an inner join on a hash computed from the concatenation of the flow ID and timestamp fields output by CICFlowMeter, applied to both the reference CSVs and the flows we wish to label. For InSDN, no ground-truth labelling beyond directory structure was provided by the dataset authors; we therefore assign labels based on the names of the directories containing the source PCAP files.

⁶ This split reflects the case of “abundant training data” in [4].

4.3 Classification Task

We frame intrusion detection as a binary classification problem, distinguishing benign flows from attack flows. All attack categories are merged into a single attack class. This applies to both $\mathcal{D}_{\text{RF}}$ and $\mathcal{D}_{\text{AE}}$, and ensures a consistent comparison between them. Flows labelled as “Attempted” attacks by Engelen *et al* [11], indicating attacks that were not executed successfully during the simulation, are considered as attack flows, as we consider that the underlying behaviour remains malicious regardless of the outcome.

The $\mathcal{D}_{\text{AE}}$ model classifies a flow as an attack if its reconstruction error exceeds a threshold. For CIC-IDS 2017, we set this threshold at the 95th percentile of the reconstruction errors observed on the training set, following Alotaibi *et al*’s work [2]. For InSDN, the scarcity of benign flows means the training set is much smaller, and applying the same 95th percentile threshold yielded an unacceptably high false positive rate. We therefore empirically explored several threshold values on the training data (as no separate calibration set was available due to data scarcity), selecting the 80th percentile as the best trade-off between true and false positive rates.

4.4 Feature Configurations

We evaluated seven feature configurations, described in Table 4, to distinguish the effect of correcting feature values from the effect of removing duplicate or redundant features. Except for φ_P , which uses the CSV files published by Liu *et al* [18], all configurations are generated with our corrected implementation and expressed using corrected feature names. For each configuration pair, the φ_i variant removes the duplicate or redundant features present in the corresponding φ'_i variant.

Configuration φ'_O reproduces the values produced by the latest CICFlowMeter version in Engelen’s repository⁵, but maps them to our corrected naming scheme. It therefore preserves the legacy values while assigning them semantically correct feature names.

Configuration φ'_S uses the feature names of the latest CICFlowMeter version in Engelen’s repository⁵, but replaces the legacy values with corrected packet and segment length values. This configuration isolates the effect of value correction without adding back previously misnamed information. Consequently, the L4 header length minimum information previously stored under `Fwd/Bwd Seg Size Min` is not preserved.

Finally, φ'_C represents the complete corrected feature set. It uses corrected packet and segment length values while preserving the previously captured L4 payload and L4 header length information under semantically explicit feature names.

4.5 Train/Test Splits

Because $\mathcal{D}_{\text{RF}}$ and $\mathcal{D}_{\text{AE}}$ rely on different training assumptions, we use model-specific train/test splitting strategies.

Table 4: Feature configurations evaluated in this work. The “Duplicates” column indicates whether duplicate or redundant length derived features are retained or removed.

Config	Duplicates	Description
φ_P	Present	Reference baseline available only for CIC-IDS 2017, taken directly from the CSV files published by Liu <i>et al.</i> [18]. It uses the original incorrect feature values and names.
φ'_O	Present	Feature set reproducing the values of flows extracted using the Engelen repository version of CICFlowMeter ⁵ , with corrected feature names. It retains legacy L4 payload length values in place of true packet lengths, along with the resulting duplicate/redundant features.
φ_O	Removed	Same as φ'_O , removing Average Packet Size , Fwd/Bwd Segment Size Avg , and Packet Length Variance .
φ'_S	Present	Feature set using the latest Engelen repository feature names ⁵ , with corrected packet and segment length values. The legacy L4 payload-length values and the L4 header-length information previously captured by Fwd/Bwd Seg Size Min are not preserved.
φ_S	Removed	Same as φ'_S , removing Packet Length Variance and Average Packet Size .
φ'_C	Present	Complete corrected feature set, preserving previously misnamed payload and header length values under explicit feature names.
φ_C	Removed	Same as φ'_C , removing Packet Length Variance and Average Packet Size .

For $\mathcal{D}_{\text{RF}}$, we first apply a stratified temporal split: for each class independently, the first 80% of flows (in their original temporal order) are assigned to training and the remaining 20% to test. This ensures all classes are represented in both splits while preserving within-class temporal ordering. The trade-off is that the global cross-class temporal interleaving present in the original data is not preserved. We then downsample the majority class to re-balance the training set (the benign class in CIC-IDS 2017, the attack class in InSDN). The test set retains its natural class distribution. Note that for InSDN, performing the temporal split before balancing could lead to the removal of rare classes from the training set; we verified that all classes remain represented after balancing.

For $\mathcal{D}_{\text{AE}}$, the training phase requires only benign flows, as the model learns to reconstruct normal traffic. For CIC-IDS 2017, similarly to [2], we use the first 693,702 benign flows of the dataset. For InSDN, where benign flows represent a small minority of the data, we use the first 90% of benign flows for training. In both cases, the remaining data constitutes the test set.

4.6 Assessment of the statistical significance of the results

Following the recommendations provided in [4], we do not limit our evaluation to a single trial per configuration. Instead, we repeat each experiment $n = 20$ times and assess the statistical significance of our results. This allows us to determine whether the difference in a specific metric between two feature configurations can be attributed to randomness. In our experiments, statistical significance is assessed using Welch’s t-test [30], with a significance threshold of $p < 0.05$. To obtain different results across experiments while preserving reproducibility, we vary the random seed r from 1 to 20.

5 Results

5.1 Random Forest

Classification Results. Table 5 addresses **RQ1** and **RQ2** for $\mathcal{D}_{\text{RF}}$. Overall, $\mathcal{D}_{\text{RF}}$ achieves very high performance across all feature configurations on both datasets, suggesting that the model is largely robust to the considered feature-set variations. With respect to **RQ1**, the corrected feature configurations do not lead to a consistent performance improvement.

On CIC-IDS 2017, φ'_O and φ_O achieve the strongest results and reproduce the performance of the published φ_P baseline, while the corrected-value configurations remain very close. With respect to **RQ2**, removing duplicate and redundant features, yielding marginal gains for some metrics in φ_S , but this pattern is not observed across all the configurations or datasets.

On InSDN, the differences are also small: φ'_O gives slightly lower accuracy, TPR, and F1 than the best-performing configurations, while the results of the remaining configurations are generally not significantly different from each other. Overall, removing duplicate or redundant features does not lead to a consistent performance change on either dataset. The complete per-class detection rates are reported in Table 7 in the appendix. They confirm the same trend: detection rates are generally stable across feature configurations, with only limited class-level differences.

Feature Importances. Figure 2 addresses **RQ3** by comparing the Gini feature importances obtained with φ'_O , φ'_C and φ'_S . The features are ordered according to their ranking in φ'_C , which we use as the reference because it is the most complete feature set while still preserving duplicate features, thereby allowing a direct comparison with the original φ'_O configuration (see Table 4). For readability, the figure only shows configurations with duplicates. Missing bars indicate that the corresponding feature is not present in the considered configuration: in φ'_O , this is due to deprecated packet- and segment-length features being replaced by corrected semantic names, whereas in φ'_S , this is due to the absence of the newly introduced segment-payload features, as described in Table 4.

Because the feature ordering is defined by φ'_C , the bars for this configuration decrease from left to right by construction. The fact that the other configurations do not follow the same decreasing pattern indicates that the fixes modify the relative importance assigned to several features. Thus, beyond their effect on classification performance, the corrected feature definitions also affect how the model behaviour is interpreted.

Most importantly, Figure 2 shows that 7 of the top-20 most important features are affected by the fixes introduced in this work. For example, **Fwd Segment Header Length Min**, previously named **Fwd Seg Size Min**, is identified as the ninth most important feature in φ'_C (top-12 in φ'_O ⁷). This shows that incorrect feature semantics can distort feature importance interpretation, and that our corrections improve interpretability.

⁷ The corresponding ranking obtained with φ'_O is reported in Figure 4 in the appendix.

Table 5: Performance of $\mathcal{D}_{\text{RF}}$ on CIC-IDS 2017 and InSDN. Values in bold are the best results and the results that are **not** significantly different from the best result ($p > 0.05$). When duplicate and no-duplicate variants differ significantly, a dagger ($\dagger$) marks the best result.

Config	Metric	CIC-IDS 2017		InSDN	
		w/ duplicates φ'_i	w/o duplicates φ_i	w/ duplicates φ'_i	w/o duplicates φ_i
φ_P	Accuracy	0.9984 $\pm$ 0.0000	—	—	—
	TPR	0.9938 $\pm$ 0.0001	—	—	—
	FPR	0.0001 $\pm$ 0.0000	—	—	—
	F1	0.9968 $\pm$ 0.0000	—	—	—
	ROC-AUC	1.0000 $\pm$ 0.0000	—	—	—
φ_O	Accuracy	0.9984 $\pm$ 0.0000	0.9984 $\pm$ 0.0000	0.9894 $\pm$ 0.0032	0.9903 $\pm$ 0.0040
	TPR	0.9938 $\pm$ 0.0001	0.9939 $\pm$ 0.0001	0.9894 $\pm$ 0.0032	0.9903 $\pm$ 0.0041
	FPR	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000
	F1	0.9968 $\pm$ 0.0000	0.9968 $\pm$ 0.0000	0.9946 $\pm$ 0.0016	0.9951 $\pm$ 0.0020
	ROC-AUC	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9999 $\pm$ 0.0000	0.9999 $\pm$ 0.0000
φ_C	Accuracy	0.9984 $\pm$ 0.0000	0.9984 $\pm$ 0.0000	0.9908 $\pm$ 0.0040	0.9901 $\pm$ 0.0037
	TPR	0.9937 $\pm$ 0.0001	0.9938 $\pm$ 0.0001	0.9907 $\pm$ 0.0041	0.9900 $\pm$ 0.0037
	FPR	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000
	F1	0.9967 $\pm$ 0.0000	0.9967 $\pm$ 0.0000	0.9953 $\pm$ 0.0020	0.9950 $\pm$ 0.0019
	ROC-AUC	1.0000 $\pm$ 0.0000[†]	1.0000 $\pm$ 0.0000	0.9999 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
φ_S	Accuracy	0.9984 $\pm$ 0.0000	0.9984 $\pm$ 0.0000 [†]	0.9906 $\pm$ 0.0037	0.9897 $\pm$ 0.0032
	TPR	0.9936 $\pm$ 0.0002	0.9938 $\pm$ 0.0001 [†]	0.9905 $\pm$ 0.0038	0.9896 $\pm$ 0.0032
	FPR	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0000	0.0001 $\pm$ 0.0001
	F1	0.9967 $\pm$ 0.0001	0.9968 $\pm$ 0.0000 [†]	0.9952 $\pm$ 0.0019	0.9948 $\pm$ 0.0016
	ROC-AUC	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9999 $\pm$ 0.0000	0.9999 $\pm$ 0.0000

Note. Apparent ties should be interpreted with caution: values are rounded for readability, whereas significance tests were computed using the unrounded scores. Therefore, when variance across runs is very small, configurations with nearly identical reported means may still differ significantly.

5.2 Anomaly Detection

Aggregated Detection Performance. Table 6 also addresses **RQ1** and **RQ2** for $\mathcal{D}_{\text{AE}}$. For CIC-IDS 2017, the results show that the configurations φ'_S and φ_S generally obtain the weakest performance as seen in Figure 3. The ROC curves also highlight that the other configurations obtain similar TPR/FPR ratio with duplicates (φ'_O and φ'_C) and without (φ_O and φ_C). These results suggests that correcting the packet and segment length values without preserving the previously captured L4 header length information removes useful information for the anomaly detector. To verify this hypothesis, we performed an additional experiment in which the **Fwd/Bwd Segment Header Length Min** features were added back to these two configurations; the improved results, reported in Table 9 in the appendix, support this explanation. To comment on **RQ2**, the effect of duplicate and redundant features removal has been studied. Our results show that duplicate features have a small impact on the overall performance in terms of true positives and false positives. However, the ROC-AUC, which provides a threshold-independent assessment of the AutoEncoder’s performance, exhibits higher value for the duplicate-preserving variants φ'_O and φ'_C with $p < 0.05$, albeit with a small numerical difference.

For InSDN, the general results are broadly comparable across configurations, but the FPR is substantially higher than for CIC-IDS 2017. This is likely due

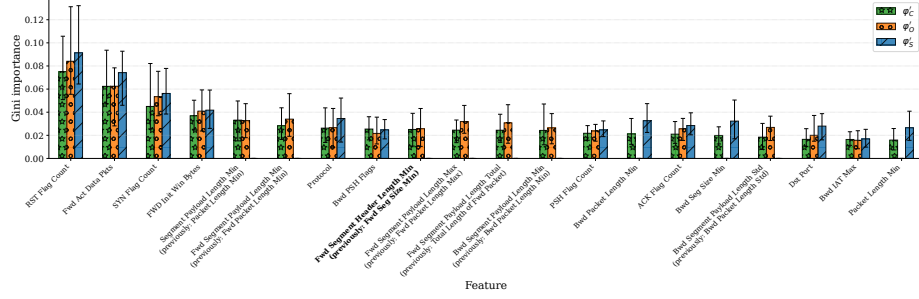


Fig. 2: Top-20 Gini feature importances for $\mathcal{D}_{RF}$ on CIC-IDS 2017, comparing ϕ'_O , ϕ'_C , and ϕ'_S . Features are ranked according to ϕ'_C . Bars report the mean importance across seeds, with whiskers spanning the minimum and maximum observed values.

to the traffic distribution in InSDN: the dataset contains far fewer benign flows (103,447) than attack flows (5,970,856), which limits the amount of benign data available to train $\mathcal{D}_{AE}$. As a result, the model has a less complete representation of normal traffic and is more likely to mislabel benign flows as anomalous.

Per-Class Detection Rates. The per-class detection rates follow the same trend as the general results. On CIC-IDS 2017, ϕ'_O often achieves the strongest detection rates (TPRs), while ϕ_S and ϕ'_S tend to perform worse overall. However, among the variants without duplicates, ϕ_C generally improves over ϕ_O , suggesting that the additional corrected length-based features partly compensate for duplicate removal, as packet, segment, and segment payload lengths capture closely related information. On InSDN, the per-class results are more clearly favourable to ϕ'_O and ϕ_O , where ϕ_S and ϕ'_S show substantially lower detection rates. The complete per-class performance overview is in Table 8 in the appendix.

6 Discussion

6.1 Stability of the performance metrics across the fixes

Despite the correction of both the values and the names of several features, neither significant improvement nor degradation were observed for $\mathcal{D}_{RF}$ or $\mathcal{D}_{AE}$. We identify two possible explanations. First, the corrected features are highly correlated with the original mislabelled ones. Packet lengths and segment lengths differ only by the IP header, whose length varies little in practice. Segment and segment payload lengths differ by the L4 header, which varies more across protocols, notably between TCP and UDP, but still represents a small part of the total length for most flows. As a result, the corrected features carry information that is close to that of the original mislabelled features, limiting the impact of the corrections on model performance.

Table 6: Performance of $\mathcal{D}_{\text{AE}}$ on CIC-IDS 2017 and InSDN. Values in bold are the best results and the results that are not significantly different from the best result. A dagger (†) marks the better value when duplicate and no-duplicate variants differ significantly ($p < 0.05$).

Config	Metric	CIC-IDS 2017		InSDN	
		w/ duplicates φ'_i	w/o duplicates φ_i	w/ duplicates φ'_i	w/o duplicates φ_i
φ_P	Accuracy	0.9196 $\pm$ 0.0122	—	—	—
	TPR	0.8674 $\pm$ 0.0331	—	—	—
	FPR	0.0506 $\pm$ 0.0015	—	—	—
	F1	0.8865 $\pm$ 0.0191	—	—	—
	ROC-AUC	0.9754 $\pm$ 0.0023	—	—	—
φ_O	Accuracy	0.9265 $\pm$ 0.0142	0.9166 $\pm$ 0.0268	0.9991 $\pm$ 0.0004	0.9656 $\pm$ 0.1499
	TPR	0.8860 $\pm$ 0.0394	0.8594 $\pm$ 0.0728	0.9994 $\pm$ 0.0004	0.9659 $\pm$ 0.1500
	FPR	0.0505 $\pm$ 0.0014	0.0507 $\pm$ 0.0012	0.3648 $\pm$ 0.0234	0.3592 $\pm$ 0.0216
	F1	0.8970 $\pm$ 0.0219	0.8806 $\pm$ 0.0434	0.9996 $\pm$ 0.0002	0.9743 $\pm$ 0.1130
	ROC-AUC	0.9759 $\pm$ 0.0032 [†]	0.9704 $\pm$ 0.0063	0.7452 $\pm$ 0.0148	0.7574 $\pm$ 0.0125 [†]
φ_C	Accuracy	0.9230 $\pm$ 0.0180	0.9143 $\pm$ 0.0192	0.9993 $\pm$ 0.0002	0.9993 $\pm$ 0.0001
	TPR	0.8766 $\pm$ 0.0491	0.8522 $\pm$ 0.0521	0.9996 $\pm$ 0.0002	0.9995 $\pm$ 0.0001
	FPR	0.0506 $\pm$ 0.0009	0.0504 $\pm$ 0.0011	0.3670 $\pm$ 0.0134	0.3609 $\pm$ 0.0177
	F1	0.8914 $\pm$ 0.0283	0.8775 $\pm$ 0.0303	0.9997 $\pm$ 0.0001	0.9996 $\pm$ 0.0001
	ROC-AUC	0.9749 $\pm$ 0.0042 [†]	0.9698 $\pm$ 0.0046	0.7572 $\pm$ 0.0104	0.7591 $\pm$ 0.0096
φ_S	Accuracy	0.9028 $\pm$ 0.0265	0.9045 $\pm$ 0.0323	0.9935 $\pm$ 0.0139	0.9904 $\pm$ 0.0225
	TPR	0.8207 $\pm$ 0.0731	0.8257 $\pm$ 0.0884	0.9938 $\pm$ 0.0139	0.9907 $\pm$ 0.0225
	FPR	0.0505 $\pm$ 0.0009	0.0506 $\pm$ 0.0014	0.3667 $\pm$ 0.0145	0.3699 $\pm$ 0.0165
	F1	0.8581 $\pm$ 0.0436	0.8602 $\pm$ 0.0552	0.9967 $\pm$ 0.0072	0.9951 $\pm$ 0.0118
	ROC-AUC	0.9697 $\pm$ 0.0063	0.9670 $\pm$ 0.0077	0.7592 $\pm$ 0.0148	0.7676 $\pm$ 0.0129

Second, both the $\mathcal{D}_{\text{RF}}$ and $\mathcal{D}_{\text{AE}}$ are indifferent to feature names: only values matter. A mislabelled feature that carries genuinely discriminative values will still contribute to model performance regardless of whether its name is semantically correct or not. Correcting the name does not change what the model learns.

6.2 Enhancement in explainability and context interpretability

Despite the absence of measurable performance improvement for $\mathcal{D}_{\text{RF}}$ and $\mathcal{D}_{\text{AE}}$, the corrections have important implications beyond raw metrics. This issue is particularly relevant for studies that interpret or analyse features. Feature-importance analyses use feature names to identify which network characteristics are useful for detecting attacks. If those names do not match the values they represent, the resulting conclusions may be incorrect. The same applies to feature-selection studies, where features are chosen or discarded based on their meaning, and to alert-explanation systems, which present feature names to analysts when explaining why a flow was classified as malicious. In all these cases, using the correct feature definitions is as important as obtaining good detection performance.

Human interpretability and XAI. Feature importance analysis is commonly used to understand model behaviour and identify which network characteristics are most relevant for detecting specific attacks. If a feature is mislabelled, any interpretation built on it is incorrect. The case of **Fwd Segment Header Length**

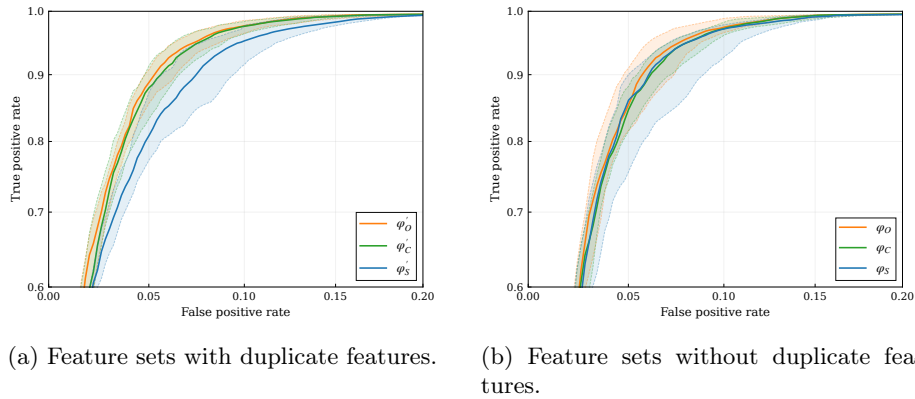


Fig. 3: ROC curves for $\mathcal{D}_{\text{AE}}$ on CIC-IDS 2017. Solid lines show the median ROC curve across seeds, and shaded regions indicate the interquartile range (Q1–Q3).

Min (previously reported as **Fwd Seg Size Min**) illustrates this concretely: researchers who relied on the original name would draw wrong/incomplete conclusions about which aspects of network traffic are most informative. As recommended by [19], understanding *why* a feature matters for a specific attack type is essential for building a trustworthy NIDS. Correct feature semantics are therefore essential for interpreting model behaviour and for making informed decisions about feature selection.

Feature correctness and LLM-based data generation. Synthetic network traffic generation is becoming increasingly important for augmenting network datasets in the intrusion detection domain [25]. Recent work has begun exploring the use of large language models for such tasks [14]. When using LLMs, which rely on feature names to understand and reproduce the statistical properties of the data, semantic correctness of feature labels is essential: a model conditioned on mislabelled features may generate traffic that is statistically plausible under the wrong interpretation, producing synthetic data that does not faithfully reflect the underlying network behaviour it is meant to represent.

6.3 Limitations

Datasets. The experiments focus on two NIDS datasets affected by the extraction flaws. While these datasets were selected because of their relevance, the analysis could be extended to datasets that capture different network scenarios, such as cloud environments (e.g., CSE-CIC-IDS-2018⁸), SCADA networks (e.g., CIC Modbus 2023⁹), or electric vehicle charging infrastructures (e.g., CIC EVSE¹⁰).

⁸ <https://www.unb.ca/cic/datasets/ids-2018.html>

⁹ <https://www.unb.ca/cic/datasets/modbus-2023.html>

¹⁰ <https://www.unb.ca/cic/datasets/evse-dataset-2024.html>

Feature importance analysis. The feature importance across the different versions of CICFlowMeter has been studied for $\mathcal{D}_{\text{RF}}$ only using the Gini importance method because of its effectiveness. Since this metric is specific to tree-based models, it cannot be applied on the $\mathcal{D}_{\text{AE}}$. For AutoEncoder-based models, other approaches such as Shapley value-based feature selection [23] could be used to analyse the importance of different features.

Impact of class imbalance. The anomaly detection experiment performed on InSDN produces poor results for the normal-class accuracy metric due to the very low volume of benign traffic. This is an inherent limitation of the dataset and could not be overcome within the scope of this work. Future research should focus on creating SDN datasets with more realistic proportions of benign and malicious traffic. Data augmentation techniques could also be considered to mitigate this problem [5].

7 Conclusion

In this work, we identified and corrected previously undocumented issues in CICFlowMeter and in datasets generated with this tool. Through experiments on CIC-IDS 2017 and InSDN, using both supervised and unsupervised machine learning-based NIDS, we showed that these corrections improve the semantic validity of the extracted features without degrading the detection performance of the tested models. The corrected features have a clear impact on feature importance analysis: several highly ranked features were affected by the fixes, showing that incorrect feature names can lead to misleading conclusions about which traffic characteristics are used by the model. These corrections therefore improve the interpretability of feature-based analyses and can support downstream tasks that rely on accurate feature semantics, such as alert explanation, feature selection, and LLM-based synthetic data generation.

We recommend future works to ensure that these fixes are applied to other datasets generated with older versions of CICFlowMeter. Additionally, our analysis of the fix impact could be extended to other NIDS models and downstream tasks, such as multi-class classification, network intrusion response, and synthetic data generation. Finally, since the strong imbalance of datasets such as InSDN limits anomaly detection experiments, future work should investigate mitigation strategies, including data augmentation and synthetic generation techniques that preserve correct feature semantics.

Disclosure of Interests. The authors declare no conflict of interest.

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Table 7: Per-class detection rates of $\mathcal{D}_{\text{RF}}$ on CIC-IDS 2017 and InSDN. Values are reported as mean $\pm$ standard deviation. WA denotes Web Attack, and † denotes “Attempted” attacks.

Attack class	φ'_O	φ_O	φ'_C	φ_C	φ'_S	φ_S
CIC-IDS 2017						
FTP-Patator †	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
FTP-Patator	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
SSH-Patator	0.9514 $\pm$ 0.0034	0.9537 $\pm$ 0.0037	0.9518 $\pm$ 0.0036	0.9530 $\pm$ 0.0037	0.9553 $\pm$ 0.0042	0.9574 $\pm$ 0.0055
SSH-Patator †	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
DoS Slowloris †	0.9912 $\pm$ 0.0012	0.9912 $\pm$ 0.0012	0.9914 $\pm$ 0.0011	0.9914 $\pm$ 0.0011	0.9849 $\pm$ 0.0289	0.9915 $\pm$ 0.0010
DoS Slowloris	0.7366 $\pm$ 0.0115	0.7388 $\pm$ 0.0121	0.7269 $\pm$ 0.0078	0.7274 $\pm$ 0.0063	0.7289 $\pm$ 0.0003	0.7289 $\pm$ 0.0000
DoS Slowhttptest	0.5647 $\pm$ 0.0044	0.5670 $\pm$ 0.0063	0.5625 $\pm$ 0.0018	0.5634 $\pm$ 0.0006	0.5632 $\pm$ 0.0000	0.5632 $\pm$ 0.0000
DoS Slowhttptest †	0.9987 $\pm$ 0.0005	0.9987 $\pm$ 0.0005	0.9986 $\pm$ 0.0003	0.9987 $\pm$ 0.0005	0.9987 $\pm$ 0.0005	0.9987 $\pm$ 0.0005
DoS Hulk	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
DoS Hulk †	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
DoS GoldenEye †	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
DoS GoldenEye	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
Heartbleed	0.9833 $\pm$ 0.0745	0.8833 $\pm$ 0.1631	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9500 $\pm$ 0.1221	0.8000 $\pm$ 0.1675
WA - Brute Force	0.9967 $\pm$ 0.0149	1.0000 $\pm$ 0.0000	0.9400 $\pm$ 0.0205	0.9600 $\pm$ 0.0335	0.9900 $\pm$ 0.0244	1.0000 $\pm$ 0.0000
WA - Brute Force †	0.9928 $\pm$ 0.0014	0.9928 $\pm$ 0.0014	0.9944 $\pm$ 0.0020	0.9948 $\pm$ 0.0019	0.9955 $\pm$ 0.0019	0.9948 $\pm$ 0.0019
WA - XSS	0.9125 $\pm$ 0.1223	0.9250 $\pm$ 0.1175	0.8000 $\pm$ 0.1026	0.8375 $\pm$ 0.1223	0.9500 $\pm$ 0.1026	0.9750 $\pm$ 0.0769
WA - XSS †	0.9885 $\pm$ 0.0039	0.9882 $\pm$ 0.0039	0.9816 $\pm$ 0.0023	0.9916 $\pm$ 0.0023	0.9924 $\pm$ 0.0000	0.9924 $\pm$ 0.0000
WA - SQL Injection †	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
WA - SQL Injection	0.6667 $\pm$ 0.0000	0.7833 $\pm$ 0.1631	0.6500 $\pm$ 0.0745	0.6833 $\pm$ 0.0745	0.7500 $\pm$ 0.1481	0.8667 $\pm$ 0.1675
Infiltration †	0.7778 $\pm$ 0.0000	0.7778 $\pm$ 0.0000	0.7778 $\pm$ 0.0000	0.7778 $\pm$ 0.0000	0.7778 $\pm$ 0.0000	0.7778 $\pm$ 0.0000
Infiltration	0.8812 $\pm$ 0.0280	0.8750 $\pm$ 0.0000	0.9250 $\pm$ 0.0628	0.9187 $\pm$ 0.0612	0.8875 $\pm$ 0.0385	0.8750 $\pm$ 0.0000
Infiltration - Portscan	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
Botnet	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
Portscan	0.9985 $\pm$ 0.0000	0.9985 $\pm$ 0.0000	0.9984 $\pm$ 0.0001	0.9984 $\pm$ 0.0001	0.9981 $\pm$ 0.0003	0.9984 $\pm$ 0.0002
Botnet †	0.7897 $\pm$ 0.0000	0.7897 $\pm$ 0.0000	0.7897 $\pm$ 0.0000	0.7897 $\pm$ 0.0000	0.7897 $\pm$ 0.0000	0.7897 $\pm$ 0.0000
DDoS	0.9989 $\pm$ 0.0000	0.9989 $\pm$ 0.0000	0.9989 $\pm$ 0.0000	0.9989 $\pm$ 0.0000	0.9989 $\pm$ 0.0000	0.9989 $\pm$ 0.0000
InSDN						
Web-attacks	0.0833 $\pm$ 0.0000	0.0833 $\pm$ 0.0000	0.0917 $\pm$ 0.0171	0.0875 $\pm$ 0.0128	0.0833 $\pm$ 0.0000	0.0875 $\pm$ 0.0128
Botnet	0.1579 $\pm$ 0.3240	0.1579 $\pm$ 0.3240	0.2421 $\pm$ 0.3684	0.2737 $\pm$ 0.3828	0.1579 $\pm$ 0.3240	0.1579 $\pm$ 0.3240
DoS	0.9066 $\pm$ 0.0032	0.9088 $\pm$ 0.0027	0.9076 $\pm$ 0.0032	0.9091 $\pm$ 0.0034	0.9182 $\pm$ 0.0089	0.9270 $\pm$ 0.0079
Probe	0.4579 $\pm$ 0.1924	0.5123 $\pm$ 0.2437	0.5400 $\pm$ 0.2443	0.4963 $\pm$ 0.2220	0.5228 $\pm$ 0.2223	0.4655 $\pm$ 0.1932
U2R	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
BFA	0.9007 $\pm$ 0.0216	0.9034 $\pm$ 0.0237	0.9128 $\pm$ 0.0383	0.9196 $\pm$ 0.0435	0.9182 $\pm$ 0.0422	0.9203 $\pm$ 0.0445
DDoS	0.9989 $\pm$ 0.0000	0.9989 $\pm$ 0.0000	0.9989 $\pm$ 0.0001	0.9989 $\pm$ 0.0001	0.9989 $\pm$ 0.0002	0.9989 $\pm$ 0.0001

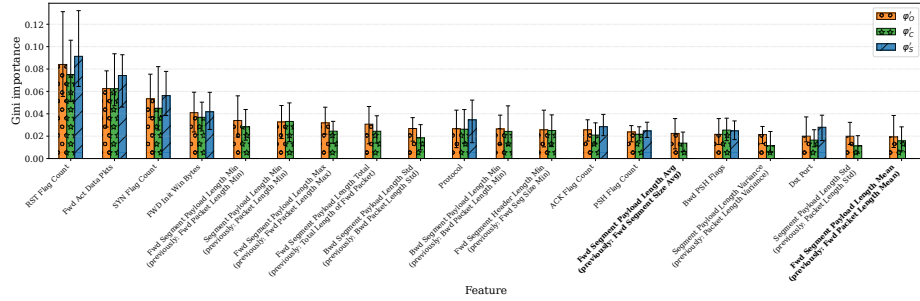


Fig. 4: Top-20 Gini feature importances of $\mathcal{D}_{\text{RF}}$ ordered according to φ'_O . Bars compare φ'_O , φ'_C , and φ'_S ; error bars indicate variability across runs. Bold labels highlight duplicate-valued forward segment payload length features that appear separately in the ranking, illustrating how duplicated features can bias the interpretation of feature importance by making the same underlying signal appear multiple times.

Table 8: Per-class detection rates of $\mathcal{D}_{AE}$ on CIC-IDS 2017 and InSDN. Values are reported as mean $\pm$ standard deviation. WA denotes Web Attack, and a dagger ($\dagger$) denotes ‘‘Attempted’’ attacks.

Attack class	φ'_O	φ_O	φ'_C	φ_C	φ'_S	φ_S
CIC-IDS 2017						
DoS Slowloris $\dagger$	0.5814 $\pm$ 0.4277	0.4499 $\pm$ 0.4513	0.9079 $\pm$ 0.0192	0.8995 $\pm$ 0.0181	0.6289 $\pm$ 0.4169	0.6310 $\pm$ 0.4223
DoS Slowloris	0.8547 $\pm$ 0.0852	0.8132 $\pm$ 0.1006	0.8950 $\pm$ 0.0758	0.8450 $\pm$ 0.1251	0.5443 $\pm$ 0.0116	0.5641 $\pm$ 0.0604
DoS Slowhttptest	0.9586 $\pm$ 0.0480	0.9491 $\pm$ 0.0453	0.9759 $\pm$ 0.0314	0.9845 $\pm$ 0.0252	0.9669 $\pm$ 0.0369	0.9585 $\pm$ 0.0365
DoS Slowhttptest $\dagger$	0.9412 $\pm$ 0.0305	0.9215 $\pm$ 0.0049	0.9901 $\pm$ 0.0054	0.9846 $\pm$ 0.0162	0.9568 $\pm$ 0.0302	0.9392 $\pm$ 0.0278
DoS Hulk	0.9897 $\pm$ 0.0134	0.9447 $\pm$ 0.0627	0.9862 $\pm$ 0.0175	0.9205 $\pm$ 0.0629	0.9955 $\pm$ 0.0046	0.9345 $\pm$ 0.0854
DoS Hulk $\dagger$	0.7406 $\pm$ 0.0584	0.7535 $\pm$ 0.0803	0.7283 $\pm$ 0.0036	0.7287 $\pm$ 0.0032	0.7215 $\pm$ 0.0042	0.7231 $\pm$ 0.0051
DoS GoldenEye	0.9434 $\pm$ 0.0233	0.8955 $\pm$ 0.0439	0.8810 $\pm$ 0.0467	0.8660 $\pm$ 0.0599	0.9333 $\pm$ 0.0315	0.9059 $\pm$ 0.0460
DoS GoldenEye $\dagger$	0.0737 $\pm$ 0.2180	0.1225 $\pm$ 0.3001	0.0250 $\pm$ 0.0000	0.0250 $\pm$ 0.0000	0.0250 $\pm$ 0.0000	0.0250 $\pm$ 0.0000
Heartbleed	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
WA - Brute Force $\dagger$	0.0014 $\pm$ 0.0003	0.0008 $\pm$ 0.0005	0.0010 $\pm$ 0.0006	0.0007 $\pm$ 0.0005	0.0013 $\pm$ 0.0005	0.0011 $\pm$ 0.0005
WA - Brute Force	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
WA - XSS $\dagger$	0.0046 $\pm$ 0.0000	0.0046 $\pm$ 0.0000	0.0046 $\pm$ 0.0000	0.0044 $\pm$ 0.0005	0.0046 $\pm$ 0.0000	0.0046 $\pm$ 0.0000
WA - XSS	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
WA - SQL Injection $\dagger$	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
WA - SQL Injection	0.0769 $\pm$ 0.0000	0.0385 $\pm$ 0.0395	0.0538 $\pm$ 0.0362	0.0231 $\pm$ 0.0362	0.0731 $\pm$ 0.0172	0.0577 $\pm$ 0.0342
Infiltration - Portscan	0.8709 $\pm$ 0.0606	0.8477 $\pm$ 0.0955	0.8888 $\pm$ 0.0610	0.8731 $\pm$ 0.0794	0.8021 $\pm$ 0.1174	0.8424 $\pm$ 0.0940
Infiltration $\dagger$	0.3633 $\pm$ 0.2674	0.3656 $\pm$ 0.2690	0.3156 $\pm$ 0.2693	0.3978 $\pm$ 0.2579	0.1822 $\pm$ 0.2533	0.0767 $\pm$ 0.1639
Infiltration	0.9639 $\pm$ 0.0131	0.9653 $\pm$ 0.0123	0.9722 $\pm$ 0.0000	0.9722 $\pm$ 0.0000	0.9625 $\pm$ 0.0136	0.9639 $\pm$ 0.0131
Botnet	0.1688 $\pm$ 0.2803	0.2141 $\pm$ 0.3389	0.2209 $\pm$ 0.3360	0.1293 $\pm$ 0.2040	0.1250 $\pm$ 0.2050	0.1227 $\pm$ 0.2056
Botnet $\dagger$	0.7908 $\pm$ 0.3905	0.7440 $\pm$ 0.4232	0.7119 $\pm$ 0.4515	0.7440 $\pm$ 0.4229	0.6639 $\pm$ 0.4699	0.5678 $\pm$ 0.4902
Portscan	0.7420 $\pm$ 0.1209	0.7265 $\pm$ 0.1645	0.7067 $\pm$ 0.1294	0.7107 $\pm$ 0.1239	0.5680 $\pm$ 0.1908	0.6395 $\pm$ 0.1959
DDoS	0.9959 $\pm$ 0.0038	0.9817 $\pm$ 0.0132	0.9953 $\pm$ 0.0048	0.9812 $\pm$ 0.0214	0.9975 $\pm$ 0.0024	0.9819 $\pm$ 0.0183
InSDN						
Web-attacks	0.9658 $\pm$ 0.0157	0.9625 $\pm$ 0.0128	0.9583 $\pm$ 0.0000	0.9583 $\pm$ 0.0000	0.7121 $\pm$ 0.1364	0.7688 $\pm$ 0.1212
Probe	0.9986 $\pm$ 0.0003	0.9970 $\pm$ 0.0069	0.9985 $\pm$ 0.0002	0.9985 $\pm$ 0.0002	0.9983 $\pm$ 0.0002	0.9984 $\pm$ 0.0004
DoS	0.9047 $\pm$ 0.0700	0.9052 $\pm$ 0.0487	0.9340 $\pm$ 0.0286	0.9264 $\pm$ 0.0185	0.4638 $\pm$ 0.1157	0.5070 $\pm$ 0.1418
Botnet	0.1269 $\pm$ 0.1981	0.0828 $\pm$ 0.0086	0.1656 $\pm$ 0.2725	0.1231 $\pm$ 0.1989	0.0780 $\pm$ 0.0059	0.0806 $\pm$ 0.0108
DDoS	1.0000 $\pm$ 0.0000	0.9657 $\pm$ 0.1534	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9968 $\pm$ 0.0141	0.9934 $\pm$ 0.0227
U2R	0.9636 $\pm$ 0.0187	0.9591 $\pm$ 0.0140	0.9545 $\pm$ 0.0000	0.9545 $\pm$ 0.0000	0.8455 $\pm$ 0.0937	0.8477 $\pm$ 0.0863
BFA	0.9720 $\pm$ 0.0856	0.9912 $\pm$ 0.0026	0.9913 $\pm$ 0.0016	0.9910 $\pm$ 0.0020	0.5311 $\pm$ 0.2219	0.5427 $\pm$ 0.1993

Note. For CIC-IDS 2017, FTP-Patator and SSH-Patator are absent from the $\mathcal{D}_{AE}$ test results because all their flows fall within the first 693,702 dataset entries used for training. Since non-benign training flows are discarded, these classes are removed from the training data and never appear in the test set.

Table 9: Performance of $\mathcal{D}_{AE}$ on CIC-IDS 2017 and InSDN when the segment header minimum features are introduced into φ_S and φ'_S . Values in bold are the best results and the results that are **not** significantly different from the best result within this restricted comparison. A dagger ($\dagger$) marks the better value when duplicate and no-duplicate variants differ significantly ($p < 0.05$).

Config	Metric	CIC-IDS 2017		InSDN	
		w/ duplicates φ'_i	w/o duplicates φ_i	w/ duplicates φ'_i	w/o duplicates φ_i
φ_S	Accuracy	0.9028 $\pm$ 0.0265	0.9045 $\pm$ 0.0323	0.9935 $\pm$ 0.0139	0.9904 $\pm$ 0.0225
	TPR	0.8207 $\pm$ 0.0731	0.8257 $\pm$ 0.0884	0.9938 $\pm$ 0.0139	0.9907 $\pm$ 0.0225
	FPR	0.0505 $\pm$ 0.0009	0.0506 $\pm$ 0.0014	0.3667 $\pm$ 0.0145	0.3699 $\pm$ 0.0165
	F1	0.8581 $\pm$ 0.0436	0.8602 $\pm$ 0.0552	0.9967 $\pm$ 0.0072	0.9951 $\pm$ 0.0118
$\varphi_S +$ Fwd/Bwd Segment Header	ROC-AUC	0.9697 $\pm$ 0.0063	0.9670 $\pm$ 0.0077	0.7592 $\pm$ 0.0148	0.7676 $\pm$ 0.0129
	Accuracy	0.9248 $\pm$ 0.0160	0.9161 $\pm$ 0.0270	0.9992 $\pm$ 0.0000	0.9992 $\pm$ 0.0000
	TPR	0.8812 $\pm$ 0.0441	0.8572 $\pm$ 0.0735	0.9995 $\pm$ 0.0000	0.9995 $\pm$ 0.0000
	FPR	0.0503 $\pm$ 0.0011	0.0503 $\pm$ 0.0012	0.3775 $\pm$ 0.0220	0.3684 $\pm$ 0.0144
Length Min	F1	0.8943 $\pm$ 0.0247	0.8797 $\pm$ 0.0436	0.9996 $\pm$ 0.0000	0.9996 $\pm$ 0.0000
	ROC-AUC	0.9759 $\pm$ 0.0043 $\dagger$	0.9709 $\pm$ 0.0066	0.7571 $\pm$ 0.0074	0.7729 $\pm$ 0.0134 $\dagger$